

Note 1.7 Introduction to Proofs

Introduction

In this section we need to learn about the notion of a proof and the methods for constructing proofs. We will know their application in both mathematics and computer science.

Some Terminology

- **Theorem:**
A statement that can be shown to be true. (also referred to as facts or results) In mathematical writing, the term theorem is somewhat important true statement.
- **Proposition:**
Less important theorems.
- **Proof:**
A valid argument that establishes the truth of a theorem.
- **Axioms(Postulates):**
Statements used in a proof we assume to be true.
- **Lemma:**
A less important theorem that is helpful in the proof of other results.
- **Corollary:**
A theorem that can be established directly from a theorem that has been proved.
- **Conjecture:**
A statement that is being proposed to be a true statement. The conjecture becomes a theorem when it is proved.

Understanding How Theorem Are Stated

Many theorems assert that a property holds for all elements in a domain, such as the integers or the real numbers. And in general the standard convention in mathematics is to omit the universal quantifiers. When proving, we often choose a general element of the domain and then use the property of the general element to make the subsequent proof.

Methods of Proving Theorems

One we have chosen a proof method, we use axioms, definitions of terms, previously proved results, and rules of inference to complete the proof.

Direct Proof

- **Definition:**

A direct proof of a conditional statement $p \rightarrow q$ is constructed by assuming p is true, then subsequent steps are constructed using rules of inference, and the final step showing q is true.

Proof by Contraposition

- **Definition:**

To prove a conditional statement $p \rightarrow q$ is true, we can prove $\neg q \rightarrow \neg p$ is true. Then take $\neg q$ as premise to obtain $\neg p$ is true.

- **Vacuous and Trivial Proofs**

Vacuous Proofs:

If we can prove the proposition p is false in the proof of $p \rightarrow q$, then we can finish the proof easily.

Trivial Proofs:

A proof of $p \rightarrow q$ that use the fact that q is true is called trivial proof.

Proofs by Contradiction

- **Definition:**

To prove a statement p is true, we can prove that $\neg p \rightarrow (r \wedge \neg r)$ is true for some proposition r . Since $r \wedge \neg r$ is F and so $\neg p$ is F. Thus p is T.

To prove a conditional statement $p \rightarrow q$ is true, we can assume p and $\neg q$ is true then we use the steps from the proof of $\neg q \rightarrow \neg p$ to show that $\neg p$ is true. Then we obtain $p \wedge \neg p$ which is F and complete the proof. That is $(p \wedge \neg q) \rightarrow F$, which is equivalent to $p \rightarrow q$.

- **Proofs of Equivalence:**

To prove a biconditional statement $p \leftrightarrow q$, we can show that $p \rightarrow q$ and $q \rightarrow p$ are both true. For a theorem states that $p_1, p_2, \dots, p_n$ are equivalent, we can use that

$$p_1 \leftrightarrow p_2 \leftrightarrow \dots \leftrightarrow p_n \leftrightarrow (p_1 \rightarrow p_2) \wedge (p_2 \rightarrow p_3) \wedge \dots \wedge (p_n \rightarrow p_1)$$

- **Counterexamples:**

To state $\forall x P(x)$ is false, we can only find a counterexample x for which $P(x)$ is false.

Mistakes in Proofs

There are many common errors made in constructing mathematical proofs.

- **Begging the question(Circular reasoning)**

The steps of a proof are based on the truth of the statement being proved or the statements equivalent to it.